

homework4

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1. Big-U notation

fibonacci.rs Find the time complexity of the different versions of the Fibonacci number computation given on ISIS with the slides from the lecture.

Version	Description	Time Complexity	Space Complexity	Notes
fib_recursive1 , fib_recursive2	Naive recursion	$O(2^n)$	$O(n)$	Exponential growth
Recursive with memoization	Cached recursion	$O(n)$	$O(n)$	Each result computed once
fib_iterative1 – fib_iterative4	Iterative / dynamic programming	$O(n)$	$O(1)$	Linear loop
fib_iterative_asm	Assembly-based iterative	$O(n)$	$O(1)$	Constant-time per iteration
fib_direct	Closed-form formula	$O(1)$	$O(1)$	Direct computation

2. Sort the sequence of functions below: (i) assuming asymptotic growth, where $g \in O(f)$ holds true for the functions g, f in a sorted sequence, and (ii) for $n = 74$ and give a short explanation.

- $4\sqrt{n}$
- $2n^3 + 4n^2 + 950$
- $\frac{n}{10000} \log_2 n$
- $\frac{1}{100} n!$
- $45 \log_2 n^2 + 9$
- 1
- $10^{110}n + 110$
- $10n + 3^n$

(i) Identify function type

- $4\sqrt{n}$: root / polynomial
- $2n^3 + 4n^2 + 950$: polynomial
- $\frac{n}{10000} \log_2 n$: $n \log n$
- $\frac{1}{100} n!$: factorial
- $45 \log_2 n^2 + 9$: logarithmic
- 1 : constant
- $10^{110}n + 110$: linear
- $10n + 3^n$: exponential

(ii) Standard asymptotic hierarchy

For large n , ignore constants :
for any c

$$1 \ll \log n \ll n^a \ll c^n \ll n!$$

so asymptotical order is :

$$1 < \log n < \sqrt{n} < n < n \log n < n^3 < 3^n < n!$$

(iii) $n = 74$

- $4\sqrt{n} = 3.440930e + 01$
- $2n^3 + 4n^2 + 950 = 8.8333020e_05$
- $\frac{n}{10000} \log_2 n = 4.594995e - 02$
- $\frac{1}{100}n! = 3.307885e + 105$
- $45 \log_2 n^2 + 9 = 5.678508e + 02$
- 1
- $10^{110}n + 110 = 7.400000e + 111$
- $10n + 3^n = 2.027556e + 35$

(iv) Order comparison

Asymptotically:

$$1 < 45 \log_2(n^2) + 9 < 4\sqrt{n} < 10^{110}n + 110 < \frac{n}{100000} \log_2 n < 2n^3 + 4n^2 + 950 < 10n + 3^n < \frac{1}{100}n!$$

For $n = 74$,

- $\frac{n}{10000} \log n$ is theoretically larger than constant 1, but due to its factor, its smaller for this n
- Linear term $10^{110}n$ is asymptotically $O(n)$, but the coefficient makes astronomically large for $n=74$

answer :

$$\frac{n}{100000} \log_2 n < 1 < 45 \log_2(n^2) + 9 < 4\sqrt{n} < 2n^3 + 4n^2 + 950 < 10n + 3^n < \frac{1}{100}n! < 10^{110}n + 110$$

(v) short explanation

With above observation, asymptotic $\neq$ numeric. asymptotic order ignores constant, while numerical comparison at a fixed n is dominated by them.